Lab 6.2 - AI Engineering Toolkit

Purpose: Consolidate only the reusable prompts, methods, agent assets, grounding checks and security controls that the learner can explain and verify.

Output: A personal classroom toolkit index plus one next-action plan.

What you will do

1. Review the outputs retained from Labs 1.1 to 6.1.
2. Select one verified RISE or E-MAP prompt.
3. Select one checked FMEA, DMAIC or SPC reference.
4. Select one tested agent configuration and its acceptance evidence.
5. Record one grounding and citation check.
6. Record the seven-step security release gate.
7. Remove sensitive, unverified or duplicate content.
8. Define one small workflow to trial after the course.

Toolkit structure

Create a folder or document with these five sections:

Section	Required content
Prompt patterns	One RISE or E-MAP prompt plus its successful correction
Engineering methods	One checked FMEA, DMAIC or SPC example with formula
Agent assets	Agent purpose, instruction, skill filename and passed tests
Grounding checks	Source scope, opened citation and missing-evidence test
Security gate	Seven evidence rows and final release-decision rule

Detailed steps

1. Gather only your own classroom outputs and the supplied fictional files.
2. For each candidate item, answer:

• What task does it support?

• What input is permitted?

• What output does it create?

• How was it checked?

- Who reviews or approves it?

3. Remove credentials, personal data, client identifiers, confidential content and unsupported claims.
4. Remove duplicate files and drafts that failed verification.
5. For the selected prompt, record the original version, the identified weakness and the corrected version.
6. For the selected engineering method, include one reproducible formula, its inputs, units and expected result.
7. For the selected agent, include its purpose, general Instructions, skill ZIP filename and acceptance-test outcomes. Do not include a live sharing link unless approved.
8. For grounding, record the exact SharePoint folder scope, one opened citation and the NOT FOUND test.
9. For security, include the seven control names and the evidence required before release.
10. Create an index with Item, Purpose, Input boundary, Verification, Owner and Location.
11. Define one next action using this structure:

```
Workflow:
Permitted input:
Required output:
Acceptance test:
Human reviewer:
Trial date:
Stop condition:
```

12. Review the toolkit with a peer or trainer and mark it **READY FOR CLASSROOM REUSE** or **REVISE**.

Pass conditions

- Every retained item has a clear purpose and verification method.
- At least one prompt, method, agent, grounding check and security gate is present.
- No credential, personal data or live confidential record remains.
- Duplicate and failed drafts are removed.
- The next action has a named reviewer and stop condition.
- Toolkit status is explicit.

This toolkit is for classroom reuse. Organisational deployment requires local governance and approval.